

Assignment-3 Question 1

1. Data Source

The data source used for this project is **NewsAPI** (<https://newsapi.org/>), which provides real-time news articles from various publishers including BBC, CNN, USA Today, and others. Articles were fetched at regular intervals using their Python API.

2. Tools and Frameworks Used

- **Apache Kafka**: For streaming real-time news data.
- **Apache Spark (Structured Streaming)**: For processing and extracting named entities.
- **spaCy**: A lightweight NLP library used for Named Entity Recognition (NER).
- **Elasticsearch**: To index and store extracted entities.
- **Kibana**: To create live visualizations (bar charts) of the top named entities.

3. Code Explanation

Producer.py

- Connects to NewsAPI and fetches recent articles.
- Sends each article title as a JSON object to Kafka topic topic1.

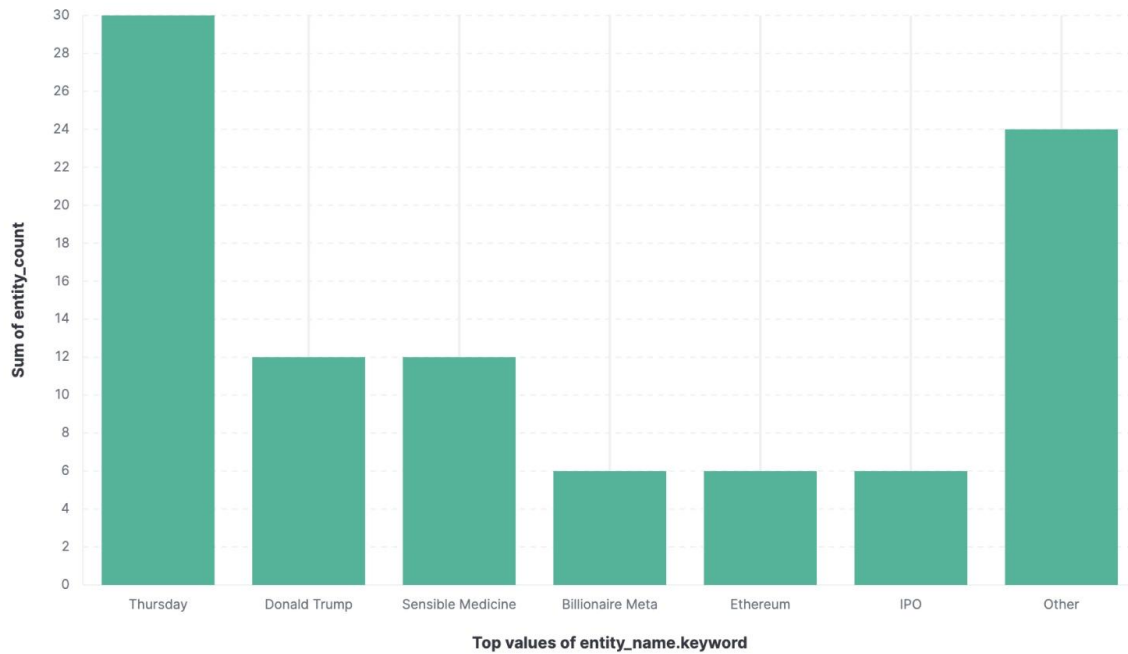
Consumer.py

- Connects to Kafka topic topic1.
- Processes incoming articles using **spaCy** to extract named entities (PERSON, ORG, GPE).
- Each extracted entity is sent to **Elasticsearch** into an index called named_entities.

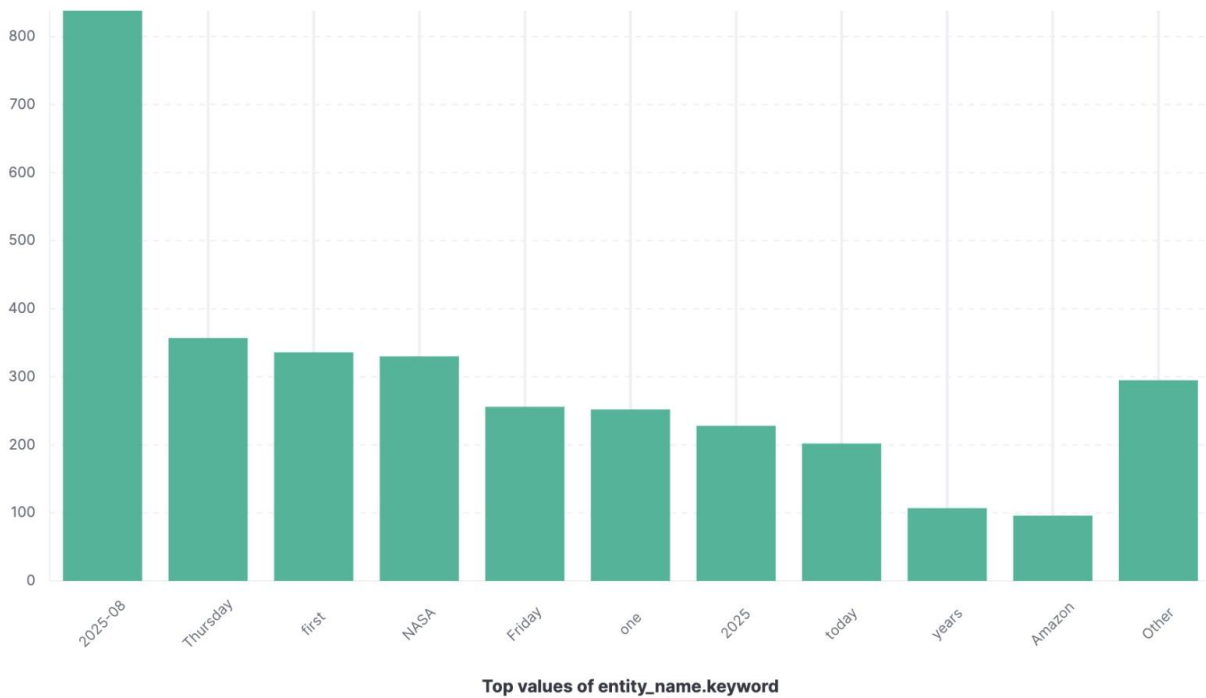
4. Dashboard Visualizations (Bar Charts)

The following bar charts were created in **Kibana**, representing the top named entities over time.

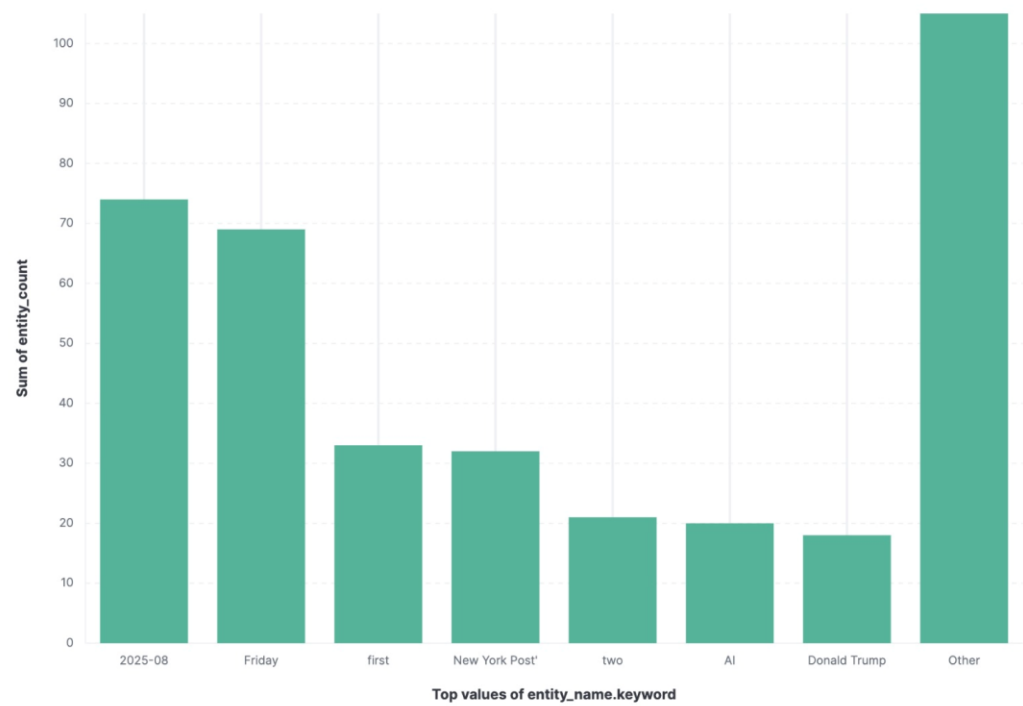
After 15 minutes:



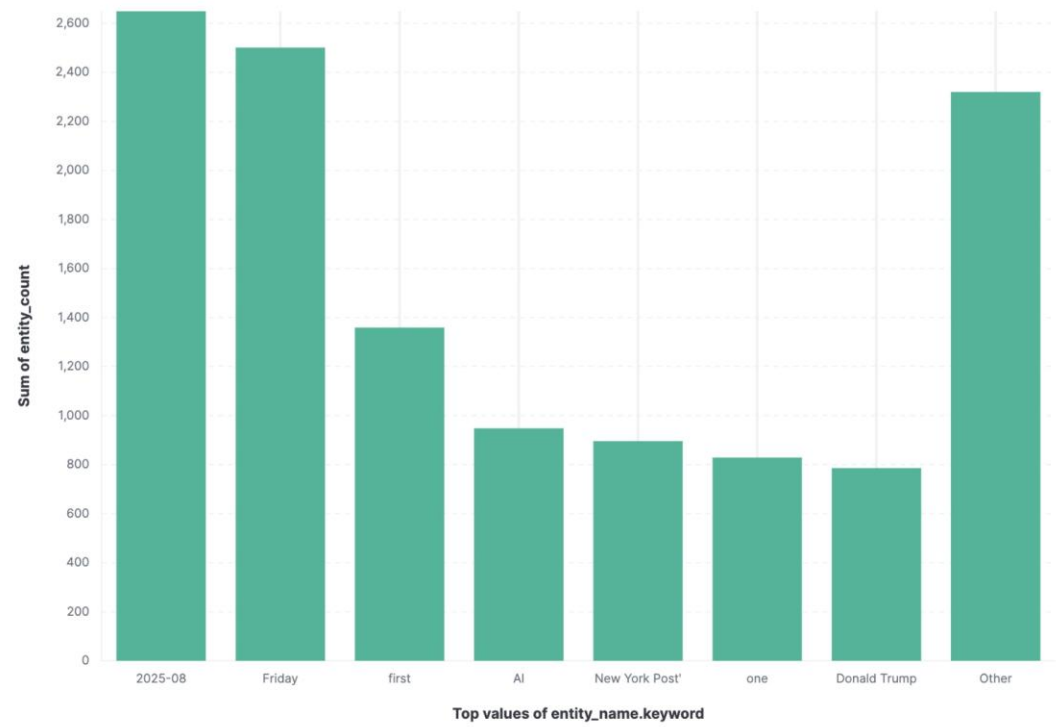
After 30 minutes:



After 45 minutes:



After 60 minutes:



Observations:

Most Mentioned Things:

- **"Thursday"** came up the most in several charts. This might mean many events or news updates happen on Thursdays.
- **"2025-08"** (August 2025) appeared often in two charts, suggesting that a lot of stories are about that month—probably about future plans or events.

Famous People and Organizations:

- **Donald Trump** was mentioned a lot, likely due to news about politics or his public appearances.
- **NASA** was also mentioned frequently, showing interest in space missions or science news.

Popular Topics:

- Words like **"Ethereum," "IPO,"** and **"Amazon"** show people are paying attention to finance, tech, and crypto.
- **"AI"** and **"New York Post"** also came up, mixing tech and media coverage.

New or Less Common Mentions:

- **"Sensible Medicine"** and **"Billionaire Meta"** were mentioned a few times, maybe showing rising interest in specific health topics or tech personalities.

Overall Patterns:

- The charts include both **time-related words** (like Thursday, August 2025) and **names** (like Trump, NASA, Amazon), showing that news covers a wide range of subjects.
- A large number of mentions were grouped under **"Other,"** which means there are many smaller topics still worth noticing.

6. Conclusion

In this project:

- Real-time news data was successfully streamed into Kafka.
- Named entities were extracted using spaCy and stored into Elasticsearch.
- Live updating dashboards were created in Kibana.

This assignment demonstrated the powerful combination of **Big Data Streaming (Kafka/Spark)** and **Real-Time Visualization (Elasticsearch/Kibana)** for extracting insights from live data sources.

All setup and code instructions have been provided in the README file separately.